

Part B — Production Architecture

Turning the little Amazon-checkout prototype into the intelligence layer of an agentic testing platform — the thing that sits between every code change and every confident release.

I wrote this the way I'd actually pitch it to a founding team: opinionated, specific enough that a senior engineer could start building from it, and honest about the parts I'd refuse to ship until I'd seen proof. Where a claim already has a working seed in Part A, I say so — §13 maps every production claim back to the function that already does a small version of it.

One framing up front, because it drives everything else. It would be easy to spend this document arguing about frameworks — LangGraph vs. a custom orchestrator, Neo4j vs. something else. I think that's the wrong fight. **The architecture that matters is the agent harness and the evaluation layer.** Get those two right and the framework underneath is a six-week refactor. Get them wrong and no framework saves you. So this document is organised around those two spines, and everything else — agents, models, graph, guardrails — hangs off them.

0 · The one belief this rests on

A browser agent and a knowledge graph have opposite strengths, and the whole design is about letting each do only what it's good at.

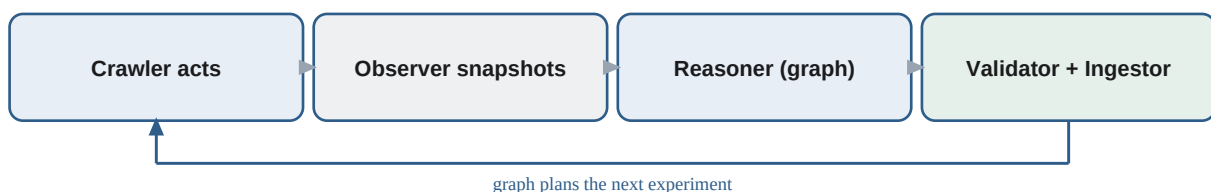
The browser agent is a **guesser**. Point it at a page and it'll find the thing that adds to cart — genuinely hard, fuzzy, page-specific work that survives Amazon shipping a redesign. But it's a terrible historian: it'll cheerfully tell you it succeeded when it didn't, and it has no memory of what it already tried.

The graph is the mirror image. It can't click anything, but it's **structural memory** — it holds everything we've ever seen about checkout and can reason over it: what's been observed, what's actually been proven, what *should* exist but doesn't.

The belief: the crawler is a probabilistic discovery device; the graph is the structural memory that tells you what discovery missed. Neither is trustworthy alone.

The value is the loop between them — and the harness that keeps that loop honest.

The failure mode I actually lose sleep over isn't the agent missing a button. You can always crawl again. It's the system **silently believing something false** — "checkout works" when the click never landed, "the promo field is gone" when the agent just wandered. So the whole architecture is built to make confident-wrong states *expensive to reach and cheap to detect*. That single idea shows up in every section below.



The core loop: the graph plans, the browser acts, the harness validates and writes memory, and the graph reasons over what it learned — then does it again.

1 · The multi-agent network

The biggest call I made is to split the work into small, single-purpose agents with hard boundaries, instead of building one clever agent that does everything. A mega-agent that decides what to test, clicks it, judges whether it worked, and writes the result is impossible to trust or debug — every failure is tangled up with every other one.

At Testsigma's scale this becomes a network of specialised agents. Here's how I'd decompose it, with the boundary each one isn't allowed to cross:

Agent	Owns	LLM?	The boundary I enforce
Planner	The frontier — what to explore/test next, priorities, budgets	No	Decides <i>what</i> ; it never touches the browser.
Crawler	One UI action via browser-use (DOM, ARIA, screenshot)	Yes (small/fast)	Executes <i>how</i> ; it never writes graph truth or decides what "passed."
Observer	Page snapshot → typed state, controls, concepts, raw artifacts	Mostly no	Hands back the <i>raw evidence</i> (DOM, screenshot, URL), not a summary to trust.
Reasoner	Missed / absent / causal scenarios from graph structure	LLM proposes, Cypher decides	The trustworthy floor is deterministic; the LLM only <i>widens</i> the candidate set.
Validator	Deterministic post-conditions after an action	No	Nothing becomes "validated" until it re-checks from independent evidence.
Healer	Selector repair when a locator breaks	Small model + graph	Proposes a fix; the Validator still has to confirm it.
Triage	Cluster failures, decide flake vs. real regression	LLM + graph history	Escalates; it doesn't silently suppress.
Safety supervisor	The guardrails, checked before every action	No	Vetoes <i>before</i> the browser ever sees the task.
Ingestor	Idempotent graph writes with provenance + confidence	No	No model calls; every write is typed and replayable.

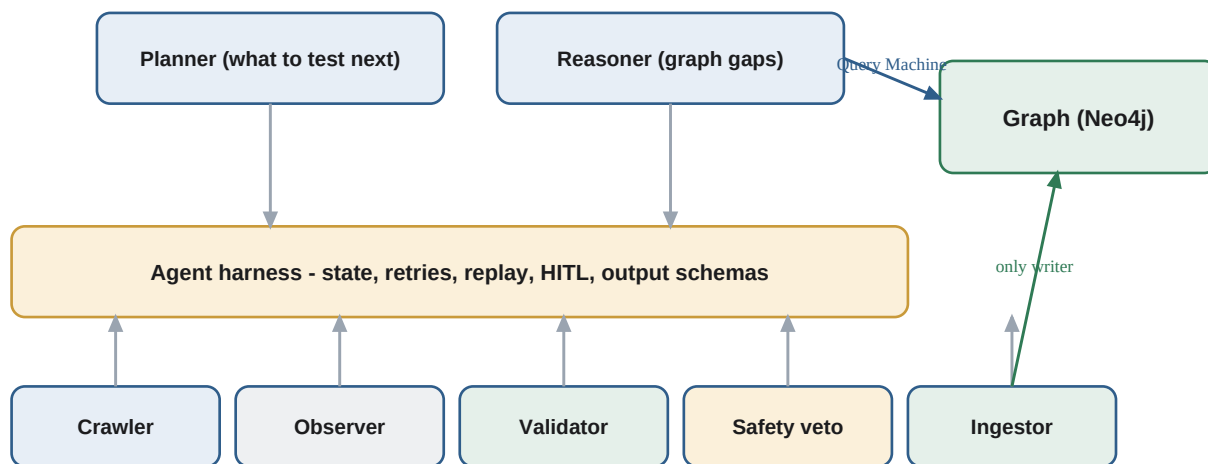
The boundary I care about most is between the Crawler and the Validator. **The Crawler is allowed to be wrong. The Validator's entire job is to not believe it.** When the Crawler says "I changed the quantity," the Validator re-reads the actual cart and checks the number moved — because the model is wrong about exactly that surprisingly often.

The other thing worth spelling out is what happens when an agent **fails, stalls, or returns low confidence** — because that's where demos and production part ways:

- **Fails** (throws, times out): the harness retries with backoff up to a budget, then marks the step failed and moves on. A failed step is data, not a crash.
- **Stalls** (keeps acting without progress — I actually hit this: the agent re-clicked "increase quantity" four times): a per-run action ledger + a no-repeat guard turns it into a fast veto instead of a runaway loop.
- **Low confidence**: the agent doesn't get to quietly pass. Below its threshold it either escalates to a stronger model, hands off to a human, or writes an explicit "unverified" marker into the graph. Uncertainty is a first-class output, not an inconvenience.

1.1 · The contract between agents

Agents don't share a blob of state — they pass **typed contracts** (Pydantic schemas in Python), and that's deliberately the most important interface in the system. Every hand-off carries: the intent, the expected state it applies to, a **confidence structure** (not a vibe — a number plus what it's conditioned on), success evidence to look for, a risk class, and an escalation signal. I'd freeze those contracts before a single agent is built, because they're what lets me swap the model or the framework underneath any one agent without the others noticing.



Only the Ingestor writes graph truth; nothing is 'validated' without the Validator.

The agent network around the harness and the graph. Nothing writes graph truth except the Ingestor, and nothing becomes "validated" without the Validator.

2 · The agent harness — this is the architecture

If I could only get one thing right, it'd be this. The **harness** is the runtime every agent runs inside. It owns state, retries, timeouts, tool dispatch, output-schema enforcement, deterministic replay, and the human-in-the-loop interrupt protocol. Frameworks like LangGraph give you the state-machine primitives; the value I'm adding lives *above* them.

What the harness owns, concretely:

- **State + replay.** Every run is a durable, resumable workflow — I'd use **Temporal** here, because crawls are long-running and I want to replay a failure step-by-step six weeks later, not guess at it. Every agent step is recorded with its inputs, outputs, model + prompt version, and a trace ID.
- **Structured tool dispatch + output validation.** Agents don't return prose that something downstream parses hopefully. They return schema-validated objects, and a response that doesn't fit the schema is a failure the harness catches — not silent garbage that becomes ground truth.
- **Retries, timeouts, circuit breakers.** Per-stage latency and token budgets. A stage that blows its budget surfaces as an observable failure, not silent spend or a hang.
- **HITL interrupts.** A clean protocol for "stop and ask a human" — with the run paused, not abandoned, so a person can answer and the workflow continues.

Here's the thing I'd say in the room and mean it: the choice between LangGraph and a custom orchestrator genuinely is a refactor. The harness contracts — *what state looks like, how replay works, how output is validated, when a human is pulled in* — are the part that's expensive to get wrong and expensive to change later. So that's what I'd design and defend first.

3 · Guardrails, designed per agent — not as global middleware

Production agent systems fail in ways demos don't, and a single global "safety middleware" is the wrong shape for it, because every agent is uncertain about different things. So guardrails are **per agent**: each one declares what "uncertain" means for it and what happens at that boundary.

The Part-A prototype already ships the sharpest example of this — the **deny-list safety veto**. I want to defend the shape of it, because it's the opposite of the obvious choice.

The tempting move is an **allow-list**: enumerate the safe actions, permit only those. Safe, and wrong for *this* system. The whole reason to put a graph behind the crawler is to surface scenarios nobody pre-conceived — an allow-list can only permit actions someone already thought of, so it would block every newly discovered or graph-inferred intent and quietly defeat the product. A guardrail that only passes what you already enumerated can't guard a system whose job is to find what you *didn't* enumerate.

So safety is a **deny-list**: default-allow, block a short, enumerable set of known-bad actions before they execute. The trick is that the genuinely dangerous actions are exactly the ones you *can* enumerate, because they're irreversible or identity-level — place order, pay, sign out, navigate off the product under test. Everything reversible (quantity, delete, save-for-later, promo) is fair game, because a mistake there is cheap and the Validator catches it.

Around that, the production guardrail layer per agent:

Guardrail	What it does
Output validators / schema enforcement	The agent's output must fit its contract or it's a failure, full stop.
Confidence gating	Below threshold → escalate, fall back, or write "unverified." No silent low-confidence passes.
Circuit breakers	An agent that fails N times in a row on a feature gets tripped and the feature is flagged, not hammered.
Fallback chains	Frontier model → small model → deterministic rule → human, in that order, per agent.
The deny-list veto	The one hard wall: irreversible actions blocked before execution, as a <i>separate</i> process a planning bug can't bypass.

4 · Model routing and composition

This role is about being the most sophisticated *consumer* of models in the room, not training them. So I'd own a routing table, and I'd defend it with eval numbers rather than brand preference. The principle is boring and correct: **deterministic by default, small models for high-volume fuzzy work, the frontier model only where reasoning value is high and volume is low.**

Task	What I'd route it to	Why
State resolution, validation, inference, ingestion	Deterministic code (free)	~90% of operations. Must be explainable and repeatable. You never use a model to check a model.
Which element to click (the Crawler)	Small/fast — GPT-4o-mini / Haiku / Gemini Flash	High volume, low stakes, latency-sensitive UI skill.
Locator-stability / flake classification	Fine-tuned small open model (Mistral-class)	Narrow, repetitive, cheap to fine-tune, and it beats a frontier model on cost at this volume.
Graph gap reasoning, root-cause chains, eval adjudication	Frontier — Claude-class, extended thinking	Low volume, high reasoning value; amortised across customers and releases.

A few opinions I'd stake:

- **Compose models without compounding hallucination.** The danger in a pipeline is one model's guess becoming the next model's "fact." The defence is the same as everywhere else here: between model stages sits a deterministic check or a graph lookup, so a guess never silently hardens into truth.

- **Pin prompt + model versions.** Every run records which model and prompt version ran.

When a provider ships a "better" model, I want to know it changed *my* decision quality before a customer does — which is a canary-eval problem (§8), not a hope.

- **Context management for long sessions.** A long agentic crawl accumulates context

fast. I'd compress aggressively: the graph *is* the long-term memory, so the working context stays small — the agent pulls exactly the graph slice it needs via the Query Machine (§6) rather than dragging the whole history along.

On cost, worked through: a feature-crawl in steady state is ~25 small-model steps at roughly 3k in / 0.3k out — call it 75k in / 7.5k out per crawl. The frontier model isn't in that path at all; it runs offline on rule discovery and eval adjudication, amortised across every customer. A customer crawling 50 features nightly is ~1,500 crawls/month, and the bill is dominated by the cheap model on purpose. The expensive levers I'd pull to keep it there: cache LLM calls by graph-state signature, invalidate bounded subgraphs instead of re-crawling everything, and never put a frontier model on the hot path.

5 · The production graph schema

The textbook hierarchy for UI testing is Element → Component → Flow → Feature. I wouldn't use it as the backbone, because it models the UI's *shape* but not the two things this platform actually sells: **provenance** (why do we believe this?) and **absence** (what should exist but doesn't?). So I built the graph around the questions we have to answer, not around the DOM tree.

Node types

Node	Key properties	A query it answers
Tenant / App / Feature	ids	Everything is scoped; one feature modeled deeply.
Run	commit_sha, started_at, model_version, cost	What did we know at run N vs. N+1?
Observation	state, url, artifact_ref, screenshot_ref	Replay exactly what the browser saw.
Concept	key, kind, observed, validated, expected, confidence, first_seen, last_seen	Expected-not-observed; observed-not-validated.
Intent	source, status, risk, selector_ref, confidence	Why an action ran, passed, or was vetoed.
Scenario	key, status, confidence, last_confirmed_run	Which scenarios are trusted right now vs. decayed.
Selector	hash, css/xpath/role, stability, last_failed	Blast radius and self-healing.
CodeArtifact	path, symbol, commit_sha	Map a PR change to selectors → concepts → scenarios.

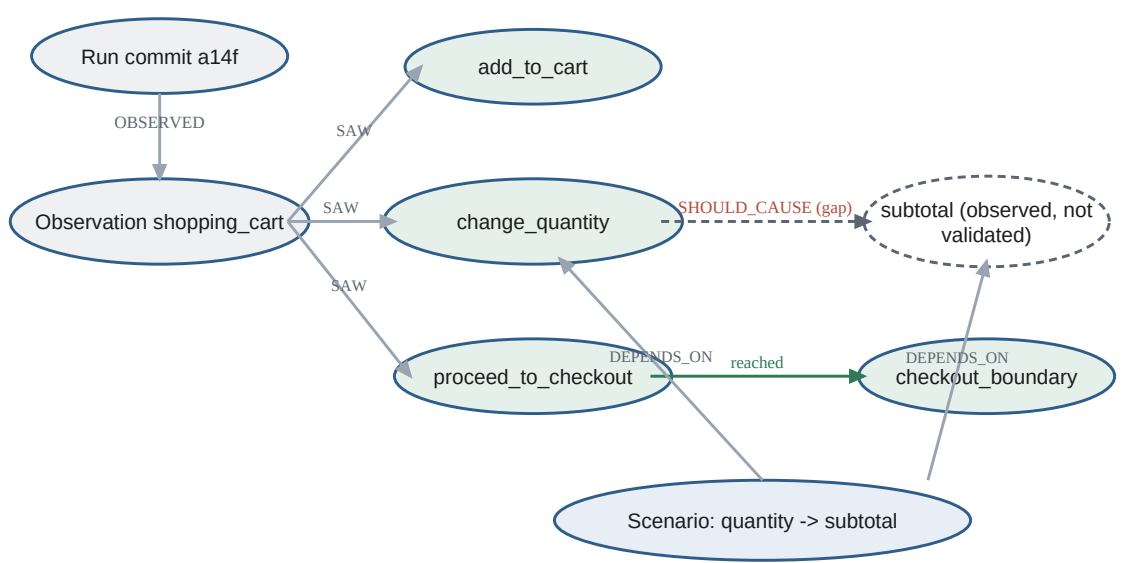
The single most important choice: **the unit of meaning is a Concept — a behaviour — not a DOM element.** Selectors are the least stable thing in the whole system; they change every redesign. Model the graph around *what a control means* ("this is the add-to-cart capability") instead of *how to find it* (#some-id), and the graph stays meaningful across redesigns while selectors churn underneath as replaceable properties.

The edges are where the value is — and I'd put **provenance and confidence on every edge**, not just nodes:

Edge	Between	What it lets me ask
OBSERVED / SAW_CONCEPT	Run → Observation → Concept	"What did we see, and where?"

TARGETS	Intent → Concept	"Which attempt exercised which behaviour, and did it pass?"
DEPENDS_ON	Scenario → Concept	"Which concepts does this scenario rely on?" (blast radius)
SHOULD_CAUSE	Concept → Concept	"This cause should produce this effect" — the causal expectation the reasoner checks
BACKS	CodeArtifact → Selector	"Which code backs this locator?" (PR risk)

Here's a concrete instance of the graph after one checkout crawl — small enough to read, real enough to reason over:



A sample knowledge graph after one Amazon-checkout crawl. Solid nodes were validated; the dashed subtotal node was observed but never proven; the dashed SHOULD_CAUSE edge is the causal gap the reasoner surfaces.

That dashed SHOULD_CAUSE edge from *change-quantity* to *subtotal* — observed cause, expected effect, never proven — is exactly the kind of missed test the graph surfaces and the crawler would never have written for itself.

5.1 · Why a graph and not just vectors

I'd push back hard on "just use RAG for this." Vector similarity is great at "find me things like this," and useless at the three things that actually matter here:

- **Absence.** "What test *should* exist and doesn't" is not a similarity query — there's nothing to be similar to. It's a structural query over expected-but-not-observed, which the graph answers in one hop and a vector store can't express at all.
 - **Temporal reasoning.** "What did the system know at commit X?" needs edges stamped with time and `commit_sha`, not nearest neighbours.
 - **Multi-hop traversal.** `CodeArtifact` → `Selector` → `Concept` → `Scenario` is a four-hop walk. Vectors give you one hop of fuzzy similarity; the walk is the whole point.
- Vectors still earn a place — selector repair and semantic matching — so I'd run **pgvector** (or Weaviate) alongside Neo4j as a **hybrid**: the graph is the source of truth, the vector index is an auxiliary lookup. It never replaces graph truth; it helps you *find* a starting node for the traversal.

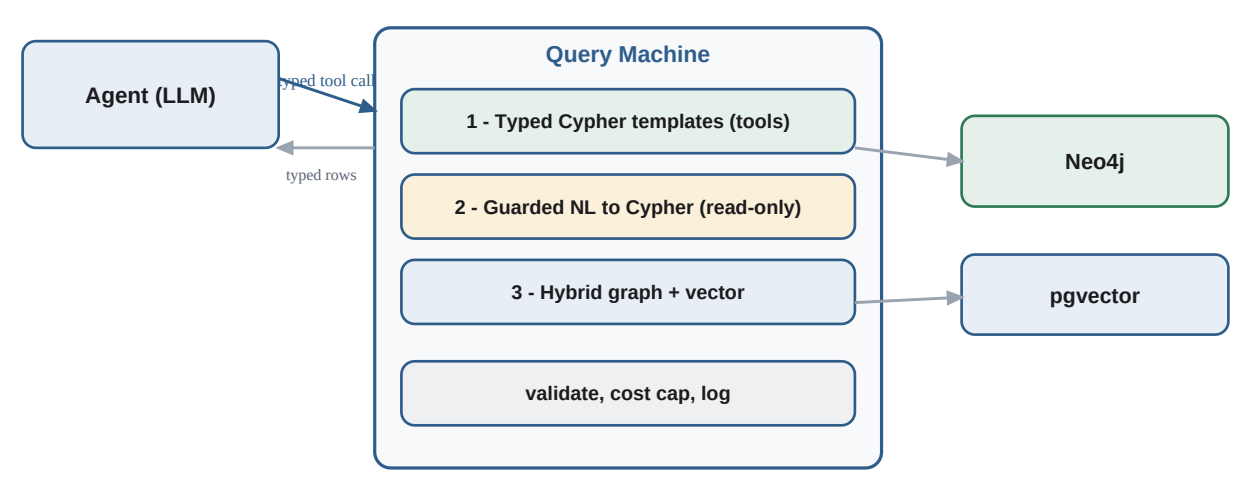
6 · The Query Machine — how the LLM actually uses the graph

A graph is only as useful as the questions the agents can ask it. Dumping the whole graph into a prompt doesn't work (context blows up, and the model drowns), and hand- writing a Cypher query for every agent need doesn't scale. So I'd build a **Query Machine**: the layer that sits between the agents and Neo4j and turns "what does the graph know?" into structured, bounded, typed answers the LLM can actually reason over.

It has three tiers, and the ordering is the whole point:

1. **A library of parameterised, typed graph queries — exposed as tools.** These are the 90% case. missed_scenarios(feature), absence(feature), blast_radius(concept), neighbours(concept, hops), confidence_of(scenario), what_did_we_know_at(commit). Each is hand-written, reviewed Cypher with a typed result schema. The agent LLM picks a query and fills the parameters as a structured tool call — it never writes Cypher. Deterministic, cacheable, can't hallucinate a traversal. 2. **Guarded natural-language → Cypher, for the long tail.** When an agent has a genuinely novel question, a frontier model translates it to Cypher — but the query runs **read-only**, against a schema allow-list, with a cost/row cap, and the generated Cypher is logged for review. If it fails validation, it falls back to tier 1. The LLM gets flexibility without getting the keys to the database. 3. **Hybrid graph + vector.** For "find the right starting node" the Query Machine can hit pgvector first (semantic match on a concept/selector), then hand the node id to a graph traversal. Similarity to *locate*, structure to *reason*.

Every answer comes back the same way: typed rows, each carrying provenance and confidence, small enough to drop into the model's context. That's what "the graph information can be fully utilised by the LLM" actually means in practice — not a bigger prompt, but a **bounded, trustworthy query interface** the agent calls like any other tool.



The Query Machine: agents ask questions as typed tool calls; deterministic Cypher templates answer the common ones; a guarded NL→Cypher path handles the long tail; every answer is typed, provenance-tagged, and small enough for the model's context.

In Part A this exists in embryo: the deterministic inference (infer_missed_scenarios) and the blast-radius traversal are the first two "tools," and the graph-expansion LLM is the first taste of the NL path. Production is mostly turning that into a real, versioned, guard-railed query library.

7 · Keeping the graph correct over time (the living property)

The brief pushes hardest here, rightly: how is the graph still correct on the 50th run, six months in, after 30 PRs? The answer is a clean split between what's incremental, recomputed, invalidated, and what compounds.

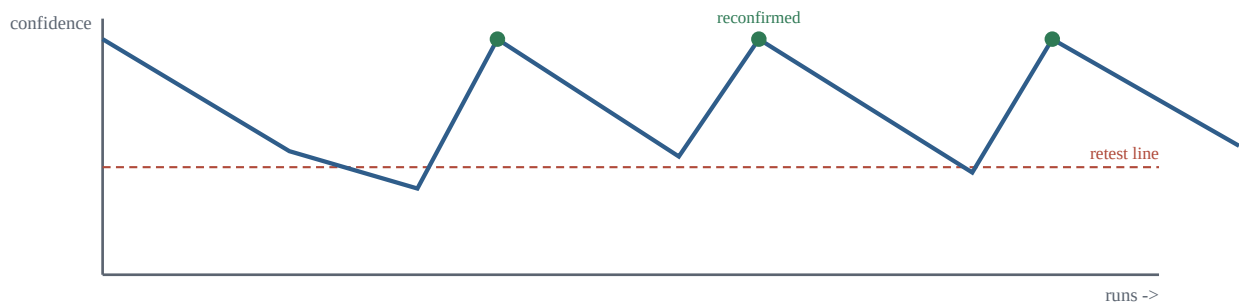
Operation	Trigger	How
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Incremental	every crawl	Upsert observations, bump <code>last_seen</code> , append evidence, add <code>Scenario-[:CONFIRMED_BY] ->Run</code> . O(seen this run).
Recomputed	a run touching a feature	Re-run inference + confidence for that feature only . Never global.
Invalidated	a PR touches mapped code; a selector fails N times; repeated validation failure	Mark a <i>bounded subgraph</i> stale; push affected scenarios into the retest frontier.
Compounds	many runs over time	Selector stability, flake rate, transition probabilities, confidence-decay curves — the six-month moat a fresh competitor can't clone.

Two things I'd design carefully because they're where "living" gets hard:

- **Conflict resolution.** When this run disagrees with last week (the promo field was here, now it's gone), I don't overwrite — I keep both, stamped with `commit_sha` and time, and let confidence decay carry the old signal down. "What did we know at time T" stays answerable because I never destroy history; I supersede it.
- **Decisions become signal.** Every validated agent decision is training signal for tomorrow — a confirmed scenario raises the prior on the next run, a repeatedly-flaky one lowers it. That's the compounding institutional asset the whole thing is about, and it's why confidence decay (below) is load-bearing rather than a nice-to-have.

Confidence decays unless something renews it — roughly, halve the distance to a floor each time a run *could* have confirmed a scenario but didn't; reset on a fresh confirmation. A scenario that drops below threshold re-enters the retest frontier on its own.



Confidence over successive runs: a scenario decays when runs don't reconfirm it, and snaps back to full confidence the moment a run proves it again. Below the retest line it re-enters the frontier automatically.

The refusal: I will not ship "recompute the whole graph nightly" as the freshness story. It's expensive and it hides drift. Bounded invalidation, proven on a golden tenant to catch a real regression, is the version I'd defend.

8 · Eval and confidence — the other spine

This is the layer that separates a research demo from a production system, and the JD is right that it *is* the architecture. Two places this system can be confidently wrong, and they need different checks: the agent clicked the wrong thing (perception), and the graph inferred a scenario that isn't real (reasoning).

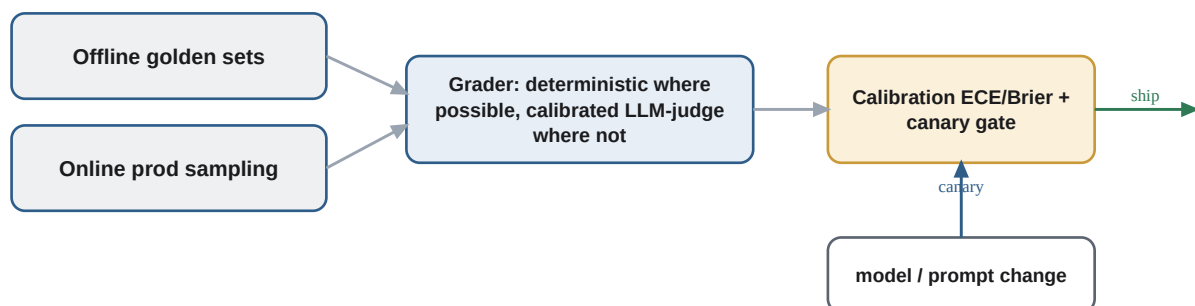
Three eval layers I'd build:

1. **Offline golden apps + regression suites.** Versioned, human-labelled trajectories. Every prompt or model change runs against them before it ships. Deterministic pass/fail plus trajectory- and step-level grading. 2. **Online production sampling.** A sampled slice of real runs graded continuously, so you catch drift the golden set didn't anticipate. 3. **Canary evals.** Every model/prompt bump runs against the golden trajectories *before* customer rollout. This is what catches a provider shipping a "better" model that quietly degrades *your* decision quality.

Question	Signal	Metric
Did it click the right target?	trace + positive/negative-target check	wrong-target rate, wander rate
Did the action succeed?	deterministic post-condition after re-observe	validation pass rate
Is the scenario reproducible?	replay across clean sessions	flake rate, pass@k
Are confidence scores honest?	predicted confidence vs. observed reproduction	ECE, Brier score
Did a model change regress us?	canary on versioned golden trajectories	delta vs. baseline

On tooling: I'd start on a managed eval store (Braintrust or LangSmith-style) to move fast, but I'd own the golden sets and the grading logic in-house, because that's the moat. **LLM-as-judge, calibrated against human spot-checks** — never an uncalibrated judge, and never a model grading something that has a deterministic check available.

The confidence point deserves emphasis because it's an architectural requirement, not a feature: **the system has to know what it doesn't know**. Every agent output carries a calibrated confidence, the harness gates on it, and I *measure* the calibration (ECE / Brier) rather than trusting it. Confidence without calibration is just false authority, and it doesn't go anywhere near a customer UI until the calibration check passes.



The eval spine: offline golden sets and online sampling feed a grader (deterministic where possible, calibrated LLM-judge where not); calibration and canary gates stand between a model change and a customer.

9 · Observability and operations

Agentic systems fail in ways ordinary services don't — same prompt, different outcome; a correct action the model *reports* as a failure; slow drift after a model upgrade. You can't operate that on plain logs.

- **A trace ID that follows a reasoning chain** across every agent call, with model + prompt version at each hop — **OpenTelemetry** as the backbone.
- **Decision audit trails** — the frontier score, the guardrail decision, the confidence, for every action that ran or was vetoed. Every decision is replayable.
- **Latency + cost budgets per agent stage**, so a stall or runaway is an observable failure, not silent spend.
- **Anomaly detection tuned for non-determinism** — the alerts I'd actually wire:

executor wander-rate climbing after a model upgrade (a canary that shipped anyway), validation-failure rate spiking on a feature (a real regression or a broken selector), stale-concept count growing without new runs (the freshness loop stalled).

And because autonomy should be *earned*, the platform moves through rollout modes rather than flipping a switch: observe-only → recommend → gated execution (safe actions auto-run, mutating ones need approval) → autonomous safe execution, only after confidence, flake rate, and safety thresholds clear.

10 · Multi-tenancy and scale

The platform has to work for a team of five and a team of five hundred, and one customer's knowledge graph must never bleed into another's. My default here is to **over-isolate**, and I'm comfortable saying so — in a tool that crawls people's apps and stores their DOM and screenshots, a cross-tenant leak is existential.

Layer	Isolated per tenant	Shared
Graph	Tenant/app/feature partition; database-per-tenant for big accounts	Schema + inference engine (code, not data)
Artifacts	Per-tenant object-store namespace, retention, redaction	Storage implementation
Models/prompts	Customer vocabulary stays isolated; versions in traces	Generic prompt templates
Learning	No selectors, DOM, screenshots, or scenarios cross tenants	Aggregated, reviewed <i>structural priors</i> only

For scale: **async agent execution** on a queue (**Celery + Redis**), long-running crawls as durable **Temporal** workflows, LLM calls cached by graph-state signature, and graph-query optimisation via the parameterised Query Machine (bounded, indexed queries instead of ad-hoc traversals). The heavy artifacts (DOM, screenshots) live in **S3/GCS**; Neo4j stores references only. Backend is **Python (async, typed) + FastAPI**, packaged in **Docker/K8s** on AWS or GCP.

Cross-customer learning is deliberately fenced. We can learn *structural priors* — "checkout flows usually have a quantity→subtotal relationship" — as reviewed, aggregated shape. We never move a selector, a DOM snapshot, or a scenario across a tenant boundary.

11 · PR blast radius

The optional PR hook isn't a separate feature — it falls straight out of the graph. A PR is a set of changed CodeArtifacts, and the answer is a downward traversal: CodeArtifact → Selector → Concept → Scenario → confidence. The discipline is in the output: it's never "rerun all the checkout tests," it's "*these* two selectors, *these* validated scenarios, and *this* one inferred scenario are at risk — retest exactly those." Part A already does the traversal (pr_blast_radius.py); production adds confidence and freshness to the ranking.

12 · What I'd refuse to ship — and the hardest problem

This is the section I'd most expect to get grilled on, which is exactly why it's here.

Refusal	Why, and what I'd need first
A path that can click final purchase — even behind a flag	Flags get flipped. Money pages must be <i>structurally</i> incapable of a final submit. First I'd want a red-team that tries to make it buy something and fails every time.
LLM self-report as validation	Every mutating action needs a deterministic post-condition. The prototype already does this for cart and checkout.

An uncalibrated confidence number in the UI	Confidence without calibration is false authority. Not until §8's calibration check passes.
Nightly full-graph recompute as the freshness story	Expensive, and it hides drift. Bounded invalidation, proven on a golden tenant.
Unreviewed cross-tenant learning	Priors may cross tenants; concepts/selectors/DOM/scenarios may not. First I'd want an audit log and a human promotion step.
The prototype's hard-coded concept vocabulary, as-is	Fine for one feature, wrong for a hundred apps. Production replaces the keyword map with a learned, per-tenant concept model behind an adapter.

And since the JD asks it directly — **the hardest unsolved problem in production agent systems, in my view: knowing a probabilistic system is wrong *before* a human does.** Everything downstream depends on confidence you can trust, and calibrating confidence is genuinely hard when your judge is itself an LLM and the distribution keeps shifting under you (new app, new model, new sprint). I don't think there's a clean solved answer. The best I have is the architecture in this document — deterministic floors under the probabilistic parts, calibrated confidence measured not assumed, canaries between a model change and a customer, and a graph that remembers what was true so you can tell when it stops being true. That's the problem I'd want to spend the next few years on.

13 · How Part A grounds every claim here

None of this is hand-waving — each production claim already has a working seed in the prototype:

Production claim	Prototype seed
Planner / Crawler split; graph plans, browser acts	<code>GraphGuidedExplorer._autonomous_loop</code> , <code>BrowserUseIntentExecutor</code>
Guardrails an LLM can't bypass (deny-list, pre-execution)	<code>safety_guard.veto_reason</code> , in the browser-use step callback
Executor output is evidence, not truth	<code>re-observe + _run_assertions</code> , <code>cart-delta / checkout verification</code>
Graph reasons about missed + causal gaps	<code>infer_missed_scenarios</code> , <code>CAUSAL_EXPECTATIONS / SHOULD_CAUSE</code>
The Query Machine (tier 1 + NL taste)	<code>infer_missed_scenarios</code> , <code>blast_radius</code> , <code>graph_expansion.expand_from_graph</code>
Absence as a graph query	<code>seed_expected_concepts</code> , <code>missing_expected_concepts</code>
Everything after the graph takes over is the graph's	<code>graph_phase_started</code> attribution
PR blast radius as a traversal	<code>pr_blast_radius.py</code> , <code>GraphStore.blast_radius</code>
Provenance + confidence on writes	<code>write_intent</code> , <code>write_observation</code> (source, confidence, last_seen)

14 · The stack, and why

Tools should fall out of the boundaries, not the other way round. Here's what I'd reach for and, just as important, what I'd avoid.

Layer	I'd start with	I'd avoid
Agent harness	Custom harness over LangGraph primitives; Temporal for durable runs	A fully autonomous agent with no state/replay contract
Browser execution	Playwright + browser-use , sandboxed	An agent with no action contract or veto

Graph store	Neo4j (temporal patterns, provenance/confidence on edges)	Graph-shaped deps in relational tables
Retrieval	pgvector / Weaviate , auxiliary — hybrid graph+vector	Letting vector similarity replace graph truth
Query Machine	Parameterised Cypher tool library + guarded NL→Cypher	Free-form model-written Cypher on the hot path
Models	Claude (frontier reasoning/judge), GPT-4o-mini / Haiku / Gemini Flash (executor), fine-tuned Mistral-class (narrow classifiers), deterministic for validation	"We use OpenAI" as the whole model strategy
Eval	Versioned golden apps; Braintrust/LangSmith-style store; calibrated LLM-judge	Shipping prompt changes without canaries
Observability	OpenTelemetry + trace IDs + decision audit	Plain logs with no trace IDs
Async / queue	Celery + Redis ; FastAPI ; Docker/K8s on AWS/GCP	Fire-and-forget cron for long, retryable crawls
Artifacts	S3 / GCS ; Neo4j holds refs only	Large DOM/screenshot blobs inside Neo4j

If I had to put the whole thing in one line: the valuable part was never the clever crawler — it's the loop between a probabilistic explorer and a structural memory, with a hard wall between what the system *did* and what it's allowed to *believe*. Get the harness and the eval layer right, and everything flashy underneath becomes a replaceable detail.